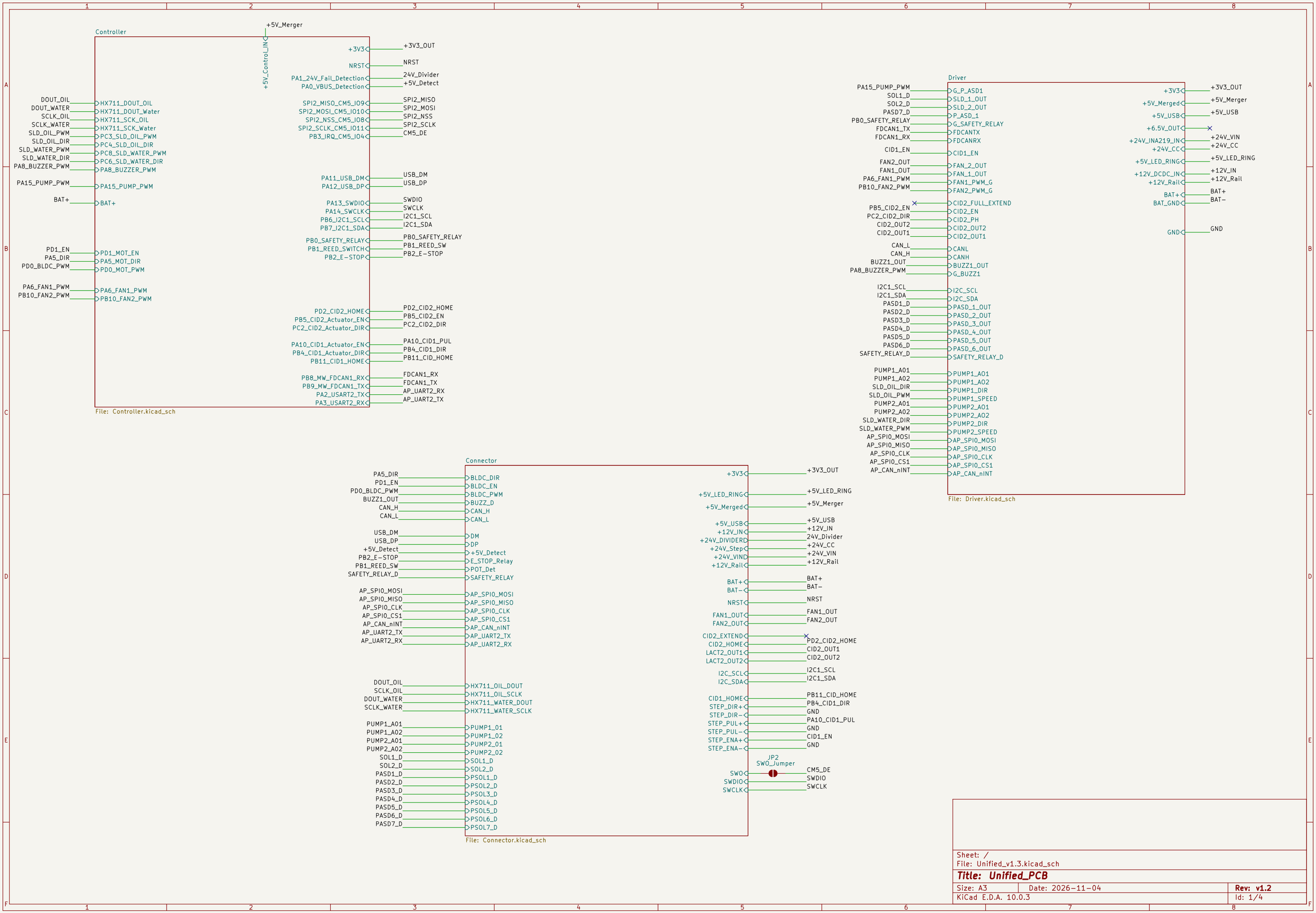
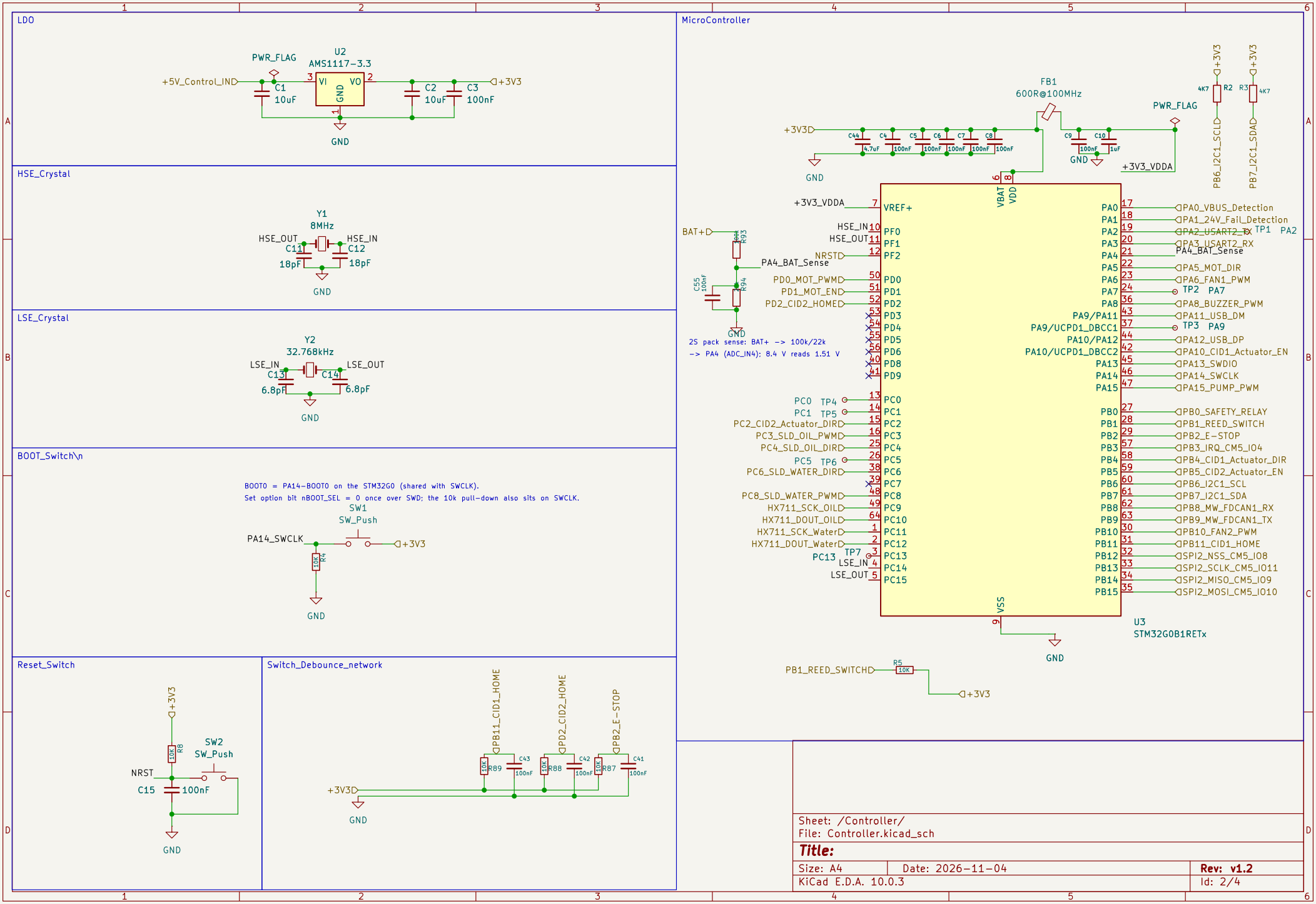
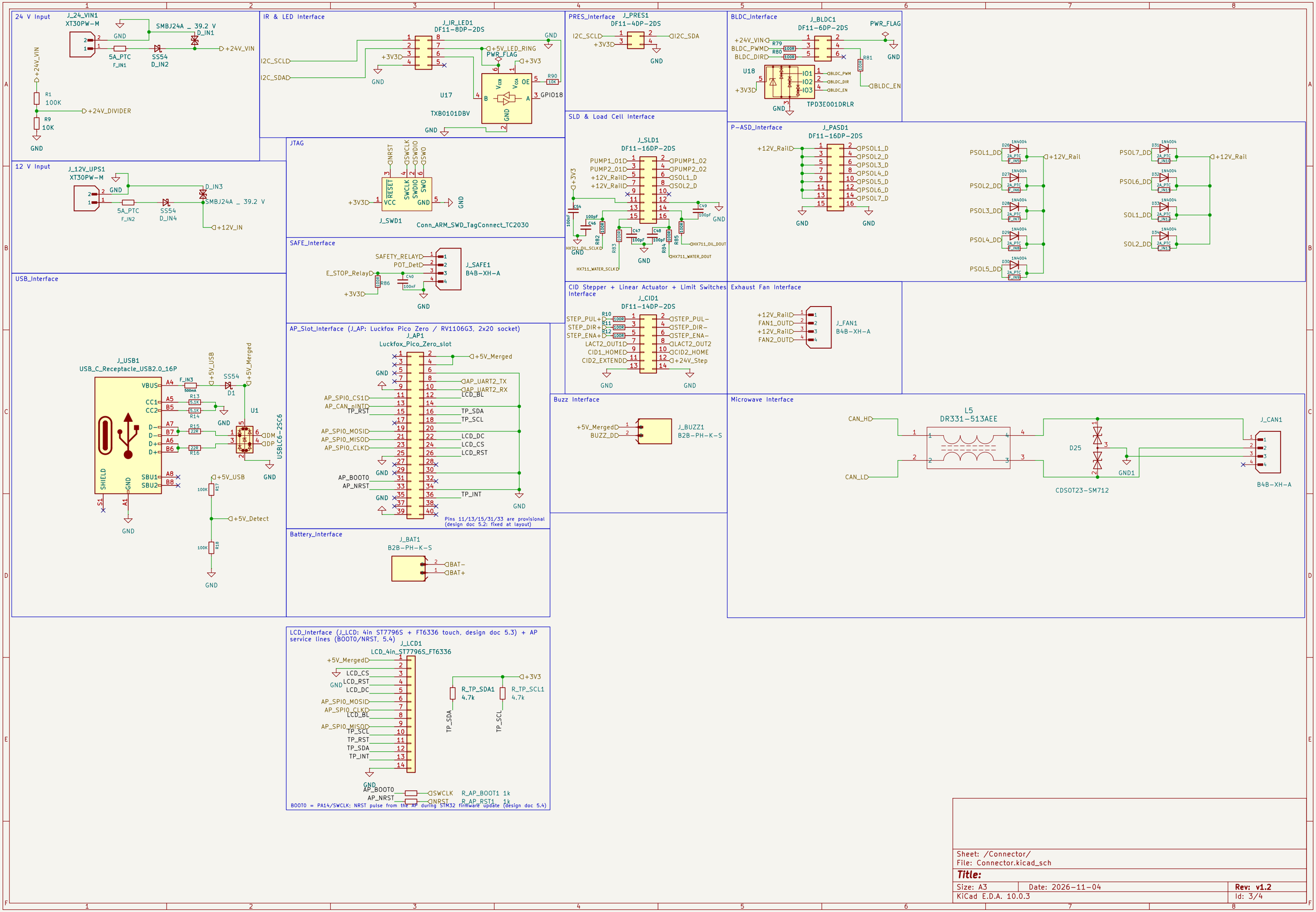


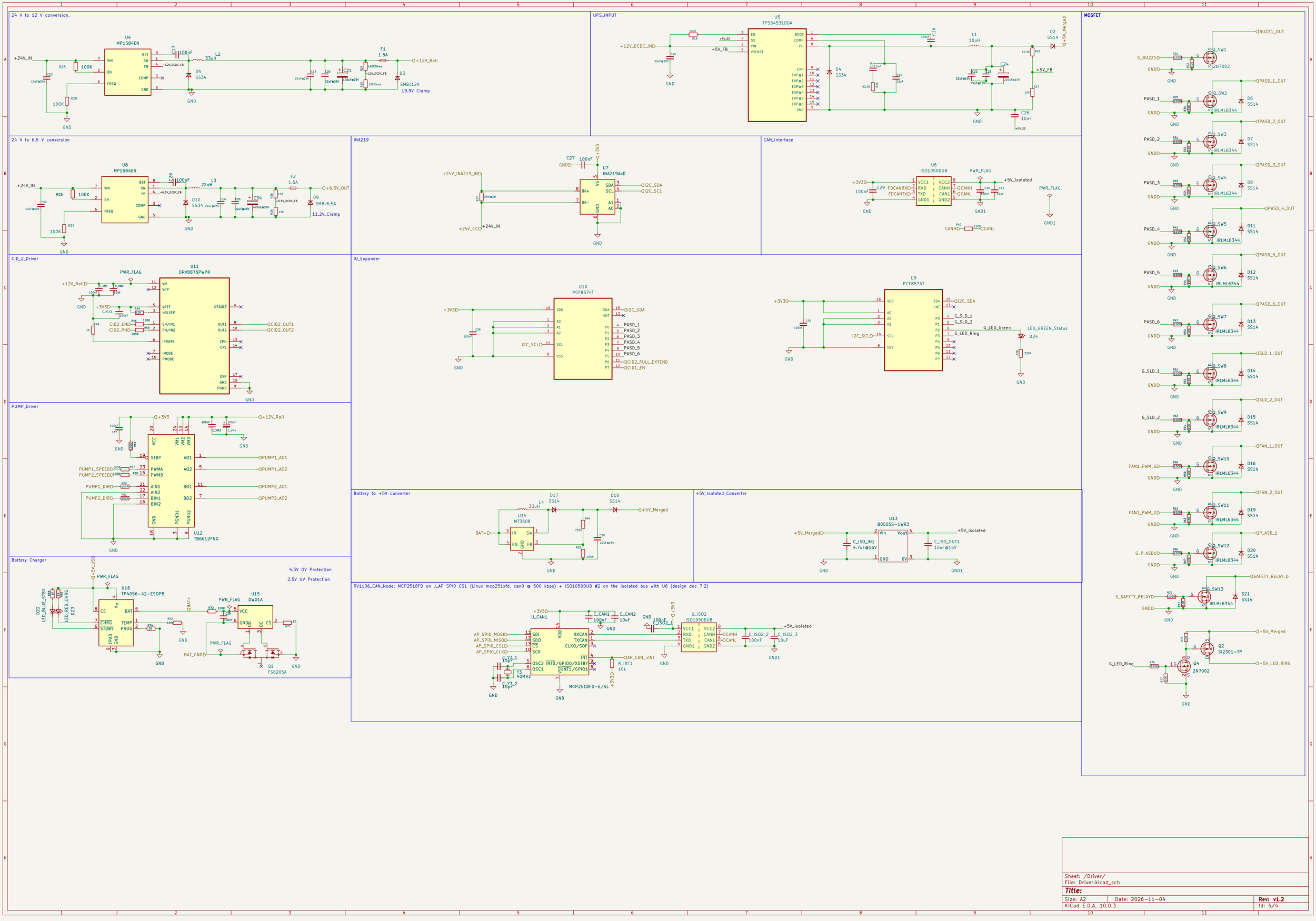
Sheet: /
File: Unified_v1.3.kicad_sch

Title: Unified_PCB

Size: A3	Date: 2026-11-04	**Rev: v1.2**
KiCad E.D.A. 10.0.3		Id: 1/4


LDO
PWR_FLAG
U2
AMS1117-3.3
+5V_Control_IND
C1
10uF
C2
10uF
C3
100nF
+3V3
GND
HSE_Crystal
Y1
8MHz
HSE_OUT
HSE_IN
C11
18pF
C12
18pF
GND
LSE_Crystal
Y2
32.768kHz
LSE_IN
LSE_OUT
C13
6.8pF
C14
6.8pF
GND
BOOT_Switch\n
BOOT0 = PA14-BOOT0 on the STM32G0 (shared with SWCLK).
Set option bit nBOOT_SEL = 0 once over SWD; the 10k pull-down also sits on SWCLK.
SW1
SW_Push
PA14_SWCLK
+3V3
R4
10K
GND
Reset_Switch
+3V3
R8
10K
SW2
SW_Push
NRST
C15
100nF
GND
Switch_Debounce_network
PB11_CID1_HOME
PD2_CID2_HOME
PB2_E-STOP
R89
10K
C43
100nF
R88
10K
C42
100nF
R87
10K
C41
100nF
+3V3
GND
MicroController
FB1
600R@100MHz
+3V3
PWR_FLAG
C44
4.7uF
C4
100nF
C5
100nF
C6
100nF
C7
100nF
C8
100nF
C9
100nF
C10
1uF
GND
+3V3_VDDA
R2
4K7
R3
4K7
PB6_I2C1_SCLD
PB7_I2C1_SDAD
VBAT
VDD
+3V3_VDDA
VREF+
BAT+
R93
4k7
C55
100nF
R94
2K
GND
2S pack sense: BAT+ -> 100k/22k
-> PA4 (ADC_IN4): 8.4 V reads 1.51 V
HSE_IN
PF0
HSE_OUT
PF1
NRST
PF2
PA4_BAT_Sense
PD0_MOT_PWM
PD1_MOT_END
PD2_CID2_HOMED
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD8
PD9
PC0 TP4
PC1 TP5
PC2_CID2_Actuator_DIR
PC3_SLD_OIL_PWMD
PC4_SLD_OIL_DIRD
PC5 TP6
PC6_SLD_WATER_DIRD
PC7
PC8_SLD_WATER_PWMD
HX711_SCK_OILD
HX711_DOUT_OILD
HX711_SCK_WaterD
HX711_DOUT_WaterD
TP7
PC13
LSE_IN
LSE_OUT
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PA0_VBUS_Detection
PA1_24V_Fail_Detection
PA2_USART2_TX
TP1 PA2
PA3_USART2_RX
PA4_BAT_Sense
PA5_MOT_DIR
PA6_FAN1_PWM
TP2 PA7
PA8_BUZZER_PWM
PA11_USB_DM
PA9/PA11
PA9/UCPD1_DBCC1
TP3 PA9
PA12_USB_DP
PA10/PA12
PA10/UCPD1_DBCC2
PA10_CID1_Actuator_EN
PA13_SWDIO
PA14_SWCLK
PA15_PUMP_PWM
PB0_SAFETY_RELAY
PB1_REED_SWITCH
PB2_E-STOP
PB3_IRQ_CM5_IO4
PB4_CID1_Actuator_DIR
PB5_CID2_Actuator_EN
PB6_I2C1_SCL
PB7_I2C1_SDA
PB8_MW_FDCAN1_RX
PB9_MW_FDCAN1_TX
PB10_FAN2_PWM
PB11_CID1_HOME
SPI2_NSS_CM5_IO8
SPI2_SCLK_CM5_IO11
SPI2_MISO_CM5_IO9
SPI2_MOSI_CM5_IO10
VSS
GND
U3
STM32G0B1RETx
PB1_REED_SWITCHD
R5
10K
+3V3
Sheet: /Controller/
File: Controller.kicad_sch
Title:
Size: A4
Date: 2026-11-04
Rev: v1.2
KiCad E.D.A. 10.0.3
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24 V Input
J_24_VIN1
XT30PW-M
SMBJ24A _ 39.2 V
D_IN1
GND
5A_PTC
F_IN1
SS54
D_IN2
+24V_VIN
+24V_VIN
R1
100K
+24V_DIVIDER
R9
10K
GND
12 V Input
J_12V_UPS1
XT30PW-M
GND
D_IN3
SMBJ24A _ 39.2 V
5A_PTC
F_IN2
SS54
D_IN4
+12V_IN
USB_Interface
J_USB1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
F_IN3
500mA
+5V_USB
SS54
D1
+5V_Merged
R13
5.1K
R14
5.1K
R15
22R
R16
22R
U1
USBLC6-2SC6
DM
DP
GND
+5V_USB
100K
R17
+5V_Detect
100K
R18
GND
IR & LED Interface
J_IR_LED1
DF11-8DP-2DS
I2C_SCL
I2C_SDA
+3V3
GND
+5V_LED_RING
PWR_FLAG
+3V3
GND
R90
10K
GPIO18
U17
TXB0101DBV
Vcca
Vccb
OE
A
B
GND
GND
JTAG
NRST
SWCLK
SWDIO
SWO
RESET
SWCLK
SWDIO
SWO
+3V3
VCC
GND
GND
J_SWD1
Conn_ARM_SWD_TagConnect_TC2030
SAFE_Interface
SAFETY_RELAY
POT_Det
E_STOP_Relay
R86
100R
C40
100nF
+3V3
GND
J_SAFE1
B4B-XH-A
AP_Slot_Interface (J_AP: Luckfox Pico Zero / RV1106G3, 2x20 socket)
J_AP1
Luckfox_Pico_Zero_slot
+5V_Merged
GND
AP_UART2_TX
AP_UART2_RX
LCD_BL
AP_SPI0_CS1
AP_CAN_nINT
TP_RST
TP_SDA
TP_SCL
AP_SPI0_MOSI
AP_SPI0_MISO
AP_SPI0_CLK
LCD_DC
LCD_CS
LCD_RST
GND
AP_BOOT0
AP_NRST
TP_INT
GND
Pins 11/13/15/31/33 are provisional (design doc 5.2: fixed at layout)
Battery_Interface
J_BAT1
B2B-PH-K-S
BAT-
BAT+
PRES_Interface
J_PRES1
DF11-4DP-2DS
I2C_SCL
I2C_SDA
+3V3
GND
SLD & Load Cell Interface
J_SLD1
DF11-16DP-2DS
PUMP1_01
PUMP2_01
+12V_Rail
+12V_Rail
PUMP1_02
PUMP2_02
SOL1_D
SOL2_D
+3V3
C54
100nF
100pF
C46
R82
100R
HX711_DIL_SCLK
HX711_WATER_SCLK
R83
100R
C47
100pF
C48
100pF
R84
100R
R85
100R
C49
100pF
GND
HX711_DIL_DOUT
HX711_WATER_DOUT
CID Stepper + Linear Actuator + Limit Switches Interface
J_CID1
DF11-14DP-2DS
STEP_PUL+
STEP_DIR+
STEP_ENA+
LACT2_OUT1
CID1_HOME
CID2_EXTEND
R10
R11
R12
100R
STEP_PUL-
STEP_DIR-
STEP_ENA-
LACT2_OUT2
CID2_HOME
+24V_Step
GND
GND
Buzz Interface
+5V_Merged
BUZZ_D
J_BUZZ1
B2B-PH-K-S
BLDC_Interface
J_BLDC1
DF11-6DP-2DS
PWR_FLAG
+24V_VIN
BLDC_PWM
BLDC_DIR
R79
100R
R80
100R
R81
100R
BLDC_EN
GND
U18
+3V3
IO1
IO2
IO3
BLDC_PWM
BLDC_DIR
BLDC_EN
TPD3E001DRLR
GND
P-ASD_Interface
J_PASD1
DF11-16DP-2DS
+12V_Rail
PSOL1_D
PSOL2_D
PSOL3_D
PSOL4_D
PSOL5_D
PSOL6_D
PSOL7_D
GND
GND
D26
D27
D28
D29
D30
D31
D32
D33
D34
1N4004
2A_PTC
F_IN5
F_IN6
F_IN7
F_IN8
F_IN9
F_IN10
F_IN11
F_IN12
F_IN13
PSOL1_DD
PSOL2_DD
PSOL3_DD
PSOL4_DD
PSOL5_DD
PSOL7_DD
PSOL6_DD
SOL1_DD
SOL2_DD
+12V_Rail
+12V_Rail
Exhaust Fan Interface
+12V_Rail
FAN1_OUT
+12V_Rail
FAN2_OUT
J_FAN1
B4B-XH-A
Microwave Interface
L5
DR331-513AEE
CAN_HD
CAN_LD
D25
CDSOT23-SM712
GND1
J_CAN1
B4B-XH-A
LCD_Interface (J_LCD: 4in ST7796S + FT6336 touch, design doc 5.3) + AP service lines (BOOT0/NRST, 5.4)
J_LCD1
LCD_4in_ST7796S_FT6336
+5V_Merged
LCD_CS
LCD_RST
LCD_DC
AP_SPI0_MOSI
AP_SPI0_CLK
LCD_BL
AP_SPI0_MISO
TP_SCL
TP_RST
TP_SDA
TP_INT
GND
+3V3
R_TP_SDA1
4.7k
R_TP_SCL1
4.7k
TP_SDA
TP_SCL
GND
AP_BOOT0
AP_NRST
SWCLK
NRST
R_AP_BOOT1 1k
R_AP_RST1 1k
BOOT0 = PA14/SWCLK; NRST pulse from the AP during STM32 firmware update (design doc 5.4)
Sheet: /Connector/
File: Connector.kicad_sch
Title:
Size: A3
Date: 2026-11-04
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Sheet: /Driver/		
File: Driver.kicad_sch		
Title:		
Size: A2	Date: 2026-11-04	**Rev: v1.2**
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